



Veni, Vidi, Vici—When Is Home Video Seizure Monitoring Helpful?

Epilepsy Currents

2025, Vol. 25(1) 9-16

© The Author(s) 2024

Article reuse guidelines:

sagepub.com/journals-permissions

DOI: 10.1177/15357597241253426

journals.sagepub.com/home/epi



Theodore A. Sheehan, BS^{1,2} , Eliza Winter-Potter, BS^{1,2},
Anna Dorste, MLIS³, Christian Meisel, MD⁴, and Tobias Loddenkemper, MD¹

¹ Division of Epilepsy and Clinical Neurophysiology, Department of Neurology, Boston Children's Hospital, Harvard Medical School, Boston, MA, USA

² Rocky Vista University College of Osteopathic Medicine, Parker, CO, USA

³ Boston Children's Hospital, Boston, MA, USA

⁴ Charité—Universitätsmedizin Berlin & Berlin Institute of Health, Berlin, Germany

*Correspondence: Tobias Loddenkemper, MD, Neurology, Harvard Medical School, Division of Epilepsy and Clinical Neurophysiology, Fegan 9, Boston Children's Hospital, 300 Longwood Avenue, Boston, MA 02115, USA.

Email: tobias.loddenkemper@childrens.harvard.edu

Abstract

Seizure detection is vital for managing epilepsy as seizures can lead to injury and even death, in addition to impacting quality of life. Prompt detection of seizures and intervention can help prevent injury and improve outcomes for individuals with epilepsy. Wearable sensors show promising results for automated detection of certain seizures, but they have limitations such as patient tolerance, impracticality for newborns, and the need for recharging. Non-contact video and audio-based technologies have become available, but a comprehensive literature review on these methods is lacking. This scoping literature review provides an overview of video and audio-based seizure detection, highlighting their potential benefits and challenges. It encompasses a thorough search and evaluation of relevant articles, summarizing methods and performances of these systems. The primary aim of this review is to examine and analyze existing research to identify patterns and gaps and establish a foundation for future advancements. We screened 7 databases using a set of standardized search criteria to minimize any potential missed articles. Four thousand four hundred eighty-seven deduplicated abstracts were screened and narrowed down to 34 studies that varied in design, algorithm methods, types of seizures detected, and performance metrics. Seizure detection sensitivity ranged from 100% to 0%, with optical flow analysis showing the highest sensitivity. The specificity of all included articles ranged from 97.7% to 60%. While limited studies reported accuracy, the highest reported was 100% using Radon Transform based technique on Dual Tree Complex Wavelet coefficients. Video and audio-based tools offer novel, noncontact approaches for detecting and monitoring seizures. Available studies are limited in sample sizes, dataset diversity, and standardized evaluation protocols, impacting the generalizability of results. Future research focusing on larger-scale investigations with diverse datasets, standardized evaluation protocols, and consistent reporting metrics is needed.

Keywords

epilepsy, artificial intelligence, algorithm

Introduction

Prompt detection of seizures and intervention can prevent injury and improve outcomes.¹ Caregivers can provide first aid and administer seizure rescue medication, reducing the risks of status epilepticus² and complications. Accurate reporting of seizures and frequency aids in treatment, risk assessment, and

potential prevention of sudden unexplained death in epilepsy (SUDEP).³ Most cases of SUDEP go unwitnessed and occur at night,⁴ so timely detection of seizures is crucial for intervention and the prevention of SUDEP.

Interest in video- and audio-based seizure detection systems is growing. These systems use algorithms to analyze recorded



Creative Commons Non Commercial CC BY-NC: This article is distributed under the terms of the Creative Commons Attribution-NonCommercial 4.0 License (<https://creativecommons.org/licenses/by-nc/4.0/>) which permits non-commercial use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the SAGE and Open Access pages (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

signals to detect real-time seizures. The algorithms extract, track, and classify the signal to determine the likelihood of a seizure. Automated techniques offer advantages over traditional methods, especially in home environments, promoting independence for individuals with epilepsy. Additionally, these systems may reduce the need for continuous supervision and expensive, time-consuming continuous EEG monitoring in hospitals. *This scoping literature review aims to provide a comprehensive overview of the current state of video- and audio-based seizure detection and provide insights into these systems' potential benefits and challenges.*

Methods

A preliminary search of “video seizure detection” in PubMed, Medline, and Cumulative Index to Nursing and Allied Health Literature was aimed to identify relevant keywords. After 2 independent reviewers evaluated these abstracts, Karayiannis et al.⁵ published one of the earliest automated video seizure detection manuscripts. Thus, we restricted publication dates to 2006 for the final search strategy. Several articles discussed automated audio-based seizure detection methods, leading us to include “audio” in our final search strategy. Keywords searched in the titles and abstracts were combined using Boolean logic with appropriate controlled vocabulary terms from relevant articles. This strategy was translated into other databases: Embase, ClinicalTrials.gov, and Web of Science. Google Scholar and MedNars were also searched to identify unpublished data and gray literature. Searches were run from 2006 to June 27, 2022, with no language filters. Validated human filters were applied where they existed.

After importing all abstracts and citations into EndNote, we deduplicated them using the method by Bramer et al.⁶ This set was then imported into Covidence systematic review software™.⁷ Two independent reviewers screened titles and abstracts for inclusion based on criteria; audio- and/or video-detection of any seizure type, patients with a diagnosis of epilepsy, and articles written in English. Exclusion criteria included animal studies, analyses solely on psychogenic nonepileptic seizures, other scoping or systematic reviews, and opinion or commentary articles. Selected citations underwent full-text assessment by the same reviewers against the inclusion and exclusion criteria. A third independent reviewer resolved all conflicts. Data from selected studies was extracted by 2 independent reviewers using Covidence. Figure 1 details the study design from database searches to final article selection. Additionally, guidelines for evaluating seizure detection devices put forth by Beniczky and Ryvlin were followed.⁸ For a detailed methodology for classifying articles into phases 0 to 4 as proposed by Beniczky and Ryvlin, see Figure 2.

Results

Search Results and Phase Classification

Our final deduplicated search yielded 4487 unique abstracts. After applying inclusion and exclusion criteria, 82 full-text

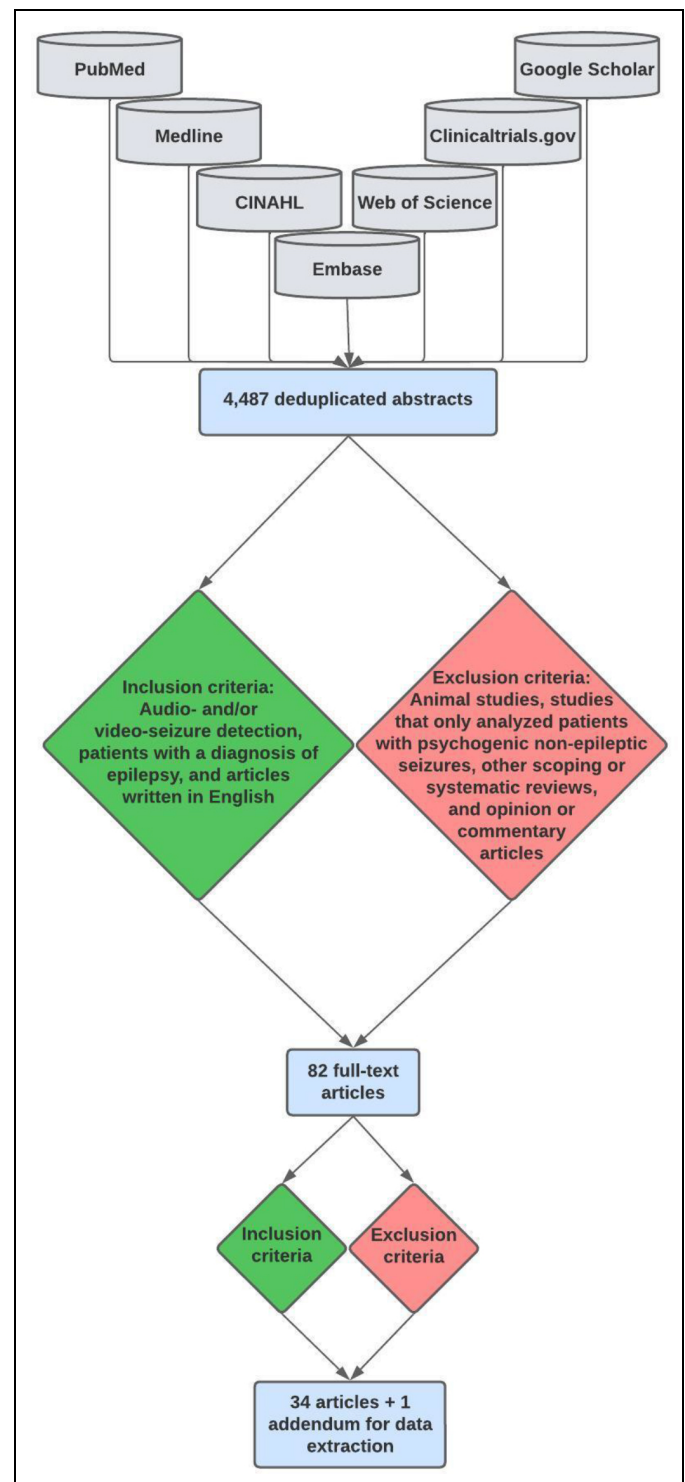


Figure 1. Methods for database searches and database sorting, including inclusion and exclusion criteria.

articles were chosen for further evaluation. Thirty-five articles, including one addendum, were then selected for extraction. A total of 34 articles underwent data extraction, and the addendum was used to update the data extracted from the original article. One study utilized audio-based seizure detection,⁹



Phase 0	Phase 1	Phase 2	Phase 3	Phase 4
Simulated seizures	< 15 real or simulated seizures from 1-10 patients	15-29 real seizures from 10-19 patients	30-74 real seizures from 20-49 patients	At least 75 real seizures from at least 50 patients
	Offline/retrospective review vEEG as reference standard	Offline/retrospective review vEEG as reference standard	Real-time continuous blinded analysis without retrospective review vEEG as reference standard	Real-time continuous blinded analysis without retrospective review vEEG as reference standard

Figure 2. Phase 0 to 4 classification.

while the remaining 33 studies were video-based seizure detection. Table 1 contains pertinent information on general study data, seizure detection methodology, and performance metrics. Many studies reported multiple values for sensitivity, specificity, and accuracy. Due to this heterogeneity, the default reporting metric in Table 1 is the mean value for these variables. When mean values were unavailable, the highest reported values were noted in Table 1.

Ten studies analyzed myoclonic seizures, 6 analyzed generalized tonic-clonic (GTC) seizures, and 15 analyzed clonic seizures. Some also assessed absence seizures and other minor seizures. The majority analyzed multiple seizure types. Only 13 of 34 studies reported recording durations, ranging from 20 minutes²⁰ to 2965 days.⁴⁰ Thirteen studies recorded data at night. Each study was reviewed to classify appropriately by phase. However, many studies exhibited characteristics of multiple phases, posing challenges in classification decisions. To address this, two reviewers (EWP and TS) independently classified each study, resolving discrepancies.

Detection Metrics for Phase 0 and Phase 1 Studies

Data on sensitivity, specificity, and accuracy is unavailable for the 3 phase 0 studies. Only 5/14 phase 1 studies reported sensitivity data. Of those, the highest sensitivity (100%) was achieved by a study that used an optical-flow-based algorithm to detect clonic seizures.²² The lowest reported sensitivity (77%) was from a study that used Harris corner detection and optical flow to detect myoclonic jerks.¹⁴ Three studies reported specificity data. The highest was 93.3% from a study that used Gaussian mixture models, coarse-to-fine paradigm, graph-cut-based segmentation, and domain knowledge to detect motor and hypermotor seizures.¹⁶ The lowest reported specificity (89%) was from a study that used an algorithm consisting of cumulative mean normalized difference function and the normalized autocorrelation function to detect clonic seizures.¹⁹ Of the 6 phase 1 studies that provided data on accuracy, the highest value (100%) was given by a study that used Radon Transform-based technique on dual tree complex wavelet Dual Tree Complex Wavelet Transform (DT-CWT).¹⁵ The lowest reported accuracy (69.8%) was provided by a study that used a leave-one-patient-out cross-validation approach.¹³

Detection Metrics for Phase 2 Studies

Eleven of 15 phase 2 studies reported data on sensitivity. Of those, the highest reported sensitivity was 100% from a study using optical flow for generalized clonic, GTC, long generalized tonic, and hyperkinetic seizures.³⁰ The lowest reported sensitivity for a phase 2 study was 70% from a region-based convolutional neural network, long short-term memory (LSTM), and leave-one-subject-out cross-validation in their specific pose analysis subanalysis.²⁷ Of the 8 studies that reported on specificity, the highest value was 97.7% using a cosine radial basis function neural network (RBFNN).⁵ The lowest value for specificity from a phase 2 study (60%) was also the same study that had the lowest sensitivity for a phase 2 study, again in their pose analysis subanalysis.²⁷ The highest reported accuracy was 95.2% from a study using an LSTM approach.⁴¹ The lowest reported accuracy was 50.9% from the same study, this time using a leave-one-subject-out cross-validation approach.⁴¹

Detection Metrics for Phase 3 and 4 Studies

Two phase 3 studies reported on sensitivity. The highest reported sensitivity (100%) was from a hybrid video/audio-based epilepsy monitoring system utilizing optical flow, specifically for detecting clonic and GTC seizures.³⁸ The lowest reported sensitivity (0%) was from the same study when they analyzed “subtle seizures.”³⁸ Only one of the two phase 3 studies reported on specificity and accuracy, with the highest reported values of 78% and 76.3%, respectively.³⁹ The single phase 4 study did not report any performance metrics.⁴⁰

Discussion

Summary

This scoping review started with 4487 unique abstracts and ultimately included 34 articles for data extraction. The review focused on audio and video-based seizure detection methods, highlighting key findings: the highest reported sensitivity was 100% in 3 studies using optical flow analysis.^{22,30,38} Specificity ranged up to 97.7% in one study that used RBFNN.⁵ Accuracy was reported by 29% of studies, with the highest at 100% using Radon Transform-based technique.¹⁵

Table 1. Main Study Results.^a

Lead author (year of publication)	Phase	Number of seizures analyzed	Seizure types included	Algorithm method	Sensitivity	Specificity	Accuracy
Kouamou (2011) ¹⁰	0	NR	Clonic, myoclonic	Template matching; optical flow; luminosity motion signal	NR	NR	NR
Lu (2011) ¹¹	0	NR	NR	Gaussian mixture models; hue saturation value modeling; graph-cut-based segmentation; coarse-to-fine detection; domain knowledge	NR	NR	NR
Murashov (2019) ¹²	0	NR	NR	Kalman-Bucy filtering algorithm; optical flow Lucas-Kanade algorithm	NR	NR	NR
Ahmed-Aristizabal (2019) ¹³	1	20	NR	3D face reconstruction model; LSTM; CNN; leave-one-patient-out cross-validation; leave-one-seizure-out cross-validation	NR	NR	Leave-one-seizure-out cross-validation: 89%; Leave-one-patient-out cross-validation 69.8%
Cuppens (2012) ¹⁴	1	14	Myoclonic jerks	Harris corner detection; optical flow; bag-of-features approach and support vector machine	77%	NR	NR
Fahmy (2018) ¹⁵	1	5	NR	Radon transform based technique on dual tree complex wavelet coefficients	NR	NR	100%
Fang (2018) ¹⁶	1	NR	NR	Spatial Temporal Gated Recurrent Unit CNN	NR	NR	NR
Lu (2013) ¹⁶	1	15	Motor, hypermotor seizures	Gaussian mixture models; coarse-to-fine paradigm; graph-cut-based segmentation; domain knowledge	Highest reported: 93.3%	Highest reported: 93.3%	NR
Mandal (2012) ¹⁷	1	2	NR	Spatio-temporal features; template matching; support vector machines; OpenCV	NR	NR	NR
Murashov (2021) ¹⁸	1	10	NR	Kalman-Bucy filtering algorithm; optical flow Lucas-Kanade algorithm	NR	NR	80%
Ntonfo (2012) ¹⁹	1	NR	Clonic	Average luminance signal; Periodicity detection algorithm comprised of cumulative mean normalized difference function and the normalized autocorrelation function	Highest reported 93%	Highest reported 89%	NR
Ntonfo (2012) ²⁰	1	NR	Clonic	Optical flow; template matching; density-based spatial clustering of applications with noise algorithm	NR	NR	NR
Ogura (2015) ²¹	1	52	Epileptic spasms	Probabilistic neural network; log-linearized Gaussian mixture network	Highest reported: 88.8% (domestic environment)	Highest reported: 93.1% (in-hospital environment)	Highest reported: 90.2% (domestic environment)
Ojanen (2021) ²²	1	36	Focal to bilateral tonic-clonic, focal tonic, focal clonic, focal myoclonic, focal motor	Semi-global matching: "PixFlow" optical flow	Clonic oscillation: 100%; tonic video + audio: 90%; tonic video 93%; tonic audio 97%	NR	NR
Pediaditis (2020) ²³	1	NR	GTC, focal to bilateral tonic-clonic, partially focal, other unspecified seizures	Dictionary learning and an autoencoder based on LSTM cells	NR	NR	85.80%
Pediaditis (2012) ²⁴	1	77	Absence, absence with myoclonic jerks	Viola-Jones detection algorithm; time-varying signal analysis; dense optical flow computation and averaging background subtraction	NR	NR	99.90%
Shen (2008) ⁹	1	NR	NR	Audio detection (hidden Markov model, Mel Frequency Cepstral Coefficient and Zero-Crossing Rate); video detection (autocorrelation of Gaussian smoothed images); combined (Dempster-Shafer fusion)	NR	NR	NR
Achilles (2018) ²⁵	2	52	Tonic, clonic, automotor	CNN	NR	NR	NR
Ahmed-Aristizabal (2019) ²⁶	2	40	NR	Optical flow; CNN; LSTM	NR	NR	Pose estimation + Optical flow accuracy: 59.3%; CNN + LSTM: 76%
Ahmed-Aristizabal (2018) ²⁷	2	161	NR	Region-based CNN; LSTM; leave-one-subject-out cross-validation scheme	Facial analysis: 85%; pose analysis 70%; hand analysis 80%	Facial analysis: 95%; pose analysis 60%; hand analysis 70%	Facial analysis: 90%; pose analysis 65%; hand analysis 75%
Ahmed-Aristizabal (2018) ²⁰	2	24	NR	Region-based CNN; LSTM; leave-one-subject-out cross-validation scheme; tasks-constrained deep convolutional network	NR	NR	LSTM: 95.2%; Leave-one-subject-out cross-validation: 50.9%
Cattani (2017) ²⁸	2	NR	Clonic	Average luminance signal; maximum-likelihood detection; Microsoft Kinect; Eulerian video magnification	Highest reported 92%	Highest reported 96%	NR

(continued)

Table 1. (continued)

Lead author (year of publication)	Phase	Number of seizures analyzed	Seizure types included	Algorithm method	Sensitivity	Specificity	Accuracy
Cattani (2014) ²⁹	2	NR	Clonic	Average luminance signal; maximum-likelihood detection; Microsoft Kinect	Highest reported: 92%	Highest reported: 92%	NR
Geertsema (2018) ³⁰	2	50	Generalized clonic, GTC, long generalized tonic, hyperkinetic	Relative frequency content in the optical flow signal	100%	NR	NR
Kalitzin (2012) ³¹	2	72	Major motor seizures	Gabor-wavelet transformations of optical flow velocities	Highest reported: 95%	NR	NR
Kalitzin (2016) ³²	2	48	Convulsive seizures	Gabor-wavelet transformations of optical flow velocities	NR	NR	NR
Karayannis (2006) ³³	2	80	Myoclonic, focal clonic	Optical flow; FFNN	Highest reported with 2 output units: 94.9%; Highest reported with 3 output units 93.9%	Highest reported with 2 output units: 86.8%; Highest reported with 3 output units 91.5%	NR
Karayannis (2006) ³⁴	2	80	Myoclonic, focal clonic	FFNN, QNN, RBFNN	Highest reported FFNN: 96.4%; QNN: 95.4%; RBFNN: 95.1%	Highest reported FFNN: 92.3%; QNN: 92.6%; RBFNN: 91.7%	NR
Karayannis (2006) ⁵	2	80	Myoclonic, focal clonic	FFNN, QNN, RBFNN	Highest reported FFNN: 97.3%; Highest reported QNN: 97.3%; Highest reported RBFNN: 99.3%	Highest reported FFNN: 94.9%; Highest reported QNN: 96.0%; Highest reported RBFNN: 97.7%	NR
vanWestrhenen (2020) ³⁵	2	230	GTC, generalized tonic lasting, focal hyperkinetic, GTC-like seizures with atypical semiology, clusters of minor seizures lasting >30 minutes	Global optical flow reconstruction iterative algorithm	78%	NR	NR
Xiong (2006) ³⁶	2	160	Myoclonic, focal clonic	Feed-forward neural networks; Pure translation block motion model; affine block motion model; fractional block motion model	Highest reported: 97.7%	Highest reported: 90.4%	NR
Yang (2021) ³⁷	2	76	GTC, focal to bilateral tonic-clonic	CNN+LSTM; leave-one-subject-out cross-validation	88%	92%	NR
Peltola (2022) ³⁸	3	48	Focal to bilateral tonic-clonic seizures, GTC, tonic, clonic, epileptic spasms, myoclonic, focal motor seizures, "subtle seizure"	Hybrid video/audio-based epilepsy monitoring system utilizing optical flow	100% for detection seizure epochs; Myoclonic: 50%; Clonic: 100%; Epileptic spasm: 27%; Tonic: 50%; GTC: 100%; Focal motor: 81.8%; "subtle seizures": 0%	NR	NR
Pisani (2014) ³⁹	3	78	Clonic	Differential average luminance	Highest reported: 86%	Highest reported: 78%	Highest reported: 76.3%
Basnyat (2022) ⁴⁰	4	2767	Unspecified, myoclonic, tonic, clonic, hyperkinetic, automatisms, tonic, clonic, GTC	Hybrid video/audio-based epilepsy monitoring system utilizing optical flow	NR	NR	NR

Abbreviations: CNN, convolutional neural network; FFNN, feedforward neural network; GTC, generalized tonic-clonic; LSTM, long short-term memory; NR, not reported; QNN, quantum neural network; RBFNN, radial basis function neural network.

*Please contact the authors for a comprehensive evaluation of video resolution, frames per second, total recording time, night recording data availability, deficiency time, false alarm rate, safety, detection latency, and user experience. Additional information regarding relevant excluded studies may also be obtained by contacting the authors.



Sensitivity

Among the studies that provided data on sensitivity, the highest reported values were 100%.^{22,30,38} All 3 studies employed optical flow analysis, effectively capturing subtle movement patterns associated with seizures, leading to high sensitivity. In one study, which uses a hybrid of audio-video data, a computer vision-based algorithm detects seizure epochs.³⁸ It is important to note that all 3 of these studies included a relatively small number of seizures: 36,²² 48,³⁸ and 50.³⁰ These smaller sample sizes may limit the representativeness and statistical power of the results compared to the study with the highest reported number of seizures analyzed (2767 seizures).⁴⁰ Although the larger study did not report sensitivity, specificity, or accuracy data, it provides a stronger foundation for drawing conclusions. Interestingly, the same study that used an audio-video hybrid detection system and reported 100% sensitivity for seizure epochs also reported the 2 lowest sensitivities of 0% and 27% for subtle seizures and epileptic spasms, respectively.³⁸ This underscores the importance of both the algorithm employed and the dataset's size and diversity. Future studies could benefit from categorizing results by seizure type for a more equitable assessment, and potentially combining separate approaches.

Specificity

A study that utilized RBFNN and included 80 myoclonic and focal clonic seizures exhibited the highest specificity of 97.7%, supporting a nonoptical flow-based approach.⁵ Data length, particularly with true negative data, greatly affects specificity estimation. Studies with sufficient nighttime data offer more true negative instances, leading to more stable specificity estimates and increased result reliability. Conversely, shorter recordings with limited true negatives may yield unreliable estimates due to sensitivity to minor variations. Proper representation of true negative data is essential for achieving a balanced assessment of a model's performance.

Accuracy

Few studies reported accuracy data. Among them, a study with a Radon Transform-based approach on DT-CWT coefficients¹⁵ exhibited the highest accuracy. However, accuracy is not always a viable metric due to the strong data imbalance, that is, many more interictal times than seizures. In such scenarios, accuracy may be misleading by simply predicting the majority class, in this case, interictal. Additionally, this study only included 5 seizures in their dataset, further limiting the interpretation of this result.

Challenges

Challenges in the review process such as search strategy, database selection, selection bias of articles, reviewer bias, and the two-reviewer approach were carefully managed as best as possible to ensure the integrity and robustness of the research. Part of our inclusion criteria was that the full texts were written in English. This may have resulted in missed pertinent articles due

to a language barrier. The first round of selection was based on abstracts and titles to limit the number of full-text reviews. Likewise, this strategy could have excluded relevant studies due to the limited scope of information initially available to the reviewers. Furthermore, it was not feasible to search every available database online, so a selection bias exists among the 7 databases we searched. To reduce reviewer bias, structured training, calibration exercises, and regular discussions were conducted. While the two-reviewer approach created a higher risk of selection bias, it also ensured a rigorous evaluation of each study, promoting consensus and reducing individual subjectivity. In split decisions, a collaborative process involving a third reviewer was employed to reach consensus, ensuring transparency, and minimizing biases.

Data collection was hindered by various reporting methods in the studies. For example, some authors aimed to collect data as mean values. However, not all studies reported means; instead, they reported the highest value for specific performance metrics. This further limits the interpretation of results because studies may have reported on outlier statistics. Data interpretation is difficult for all studies without true mean values and standard deviations.

Sample sizes varied widely across studies, with datasets often tailored to specific patient populations or controlled conditions. Understanding outcomes was influenced by the type of seizure detected, notably highlighting challenges with absence seizures compared to visually evident GTC seizures. Some studies focused on specific seizure types or detected a combination, while others did not specify seizure types. One study achieved exceptional accuracy at 99.9%,²⁴ specializing in absence seizure detection. However, lacking sensitivity and specificity data restricted result interpretation.

Additionally, the lack of a standardized evaluation protocol makes directly comparing the results across studies challenging. Future research should address these limitations by conducting large-scale studies with consistent statistical approaches for comparison. There is also a need for standardized datasets for covalidation and to exchange algorithms to identify the best-performing approaches.

Conclusion

This review showcases the potential of video and audio-based seizure detection, emphasizing diverse high-performing methodologies. However, limited sample sizes and a lack of standardized evaluation protocols indicate the necessity for more research. Consistent reporting metrics are vital for evaluating these systems. Improvements in testing and algorithms will advance noninvasive seizure detection, benefiting epilepsy management. Integrating advanced machine learning and AI techniques can enhance accuracy, reliability, and pave the way for personalized treatment strategies. Like the historical use of “veni, vidi, vici,” automated and AI powered video detection of seizures is here to stay and slowly but surely cementing its position as a feasibly deployed tool in clinical seizure detection and classification.

Authors' Note

TAS and EWP contributed equally to this manuscript. The authors confirm that they have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

Acknowledgments

The authors would like to acknowledge Keunyoung Bae for her contributions to abstract screening.

Author Contributions

TS and EWP participated in drafting and revising the manuscript for content, including medical writing, in study concept and design, data acquisition, analysis and interpretation of data, statistical analysis, and study supervision or coordination. AD, CM, and TL participated in revising the manuscript for content, in study concept and design, analysis and interpretation of data, and study supervision or coordination. All authors approved the final manuscript as submitted and agreed to be accountable for all aspects of the work.

Declaration of Conflicting Interests

The author(s) declared the following potential conflicts of interest with respect to the research, authorship, and/or publication of this article. TL is part of pending patent applications to detect and predict seizures and diagnose epilepsy. He receives research support from the NIH, the Epilepsy Research Fund, the Epilepsy Foundation of America Epilepsy Venture Fund (Shark Tank Award), MIKU, and Epitel. He received device donations from Epitel, MIKU, Empatica, and neuroelectrics. The remaining authors have no disclosures relevant to this manuscript.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This study required no funding, and no funding was received for this work. TL was supported by the Epilepsy Research Fund. TAS was supported by the Rocky Vista University Physician Scientist Track.

ORCID iD

Theodore A. Sheehan  <https://orcid.org/0000-0002-8059-9388>

References

1. Fiest KM, Sauro KM, Wiebe S, et al. Prevalence and incidence of epilepsy: a systematic review and meta-analysis of international studies. *Neurology*. 2017;88(3):296-303. doi:10.1212/WNL.0000000000003509. Erratum in: *Neurology*. 2017;89(6):642.
2. Chin RFM, Neville BGR, Peckham C, et al., NLSTEPSS Collaborative Group. Incidence, cause, and short-term outcome of convulsive status epilepticus in childhood: prospective population-based study. *Lancet*. 2006;368(9531):222-229. doi:10.1016/S0140-6736(06)69043-0
3. Thurman DJ, Hesdorffer DC, French JA. Sudden unexpected death in epilepsy: assessing the public health burden. *Epilepsia*. 2014;55(10):1479-1485. doi:10.1111/epi.12666
4. Papacostas SS. Sudden unexpected death in epilepsy: experience from a tertiary epilepsy centre in Cyprus with review of the literature. *Hippokratia*. 2015;19(4):338-343.
5. Karayiannis NB, Xiong Y, Tao G, et al. Automated detection of videotaped neonatal seizures of epileptic origin. *Epilepsia*. 2006;47(6):966-980. doi:10.1111/j.1528-1167.2006.00571.x
6. Bramer WM, Giustini D, de Jonge GB, Holland L, Bekhuis T. De-duplication of database search results for systematic reviews in EndNote. *J Med Libr Assoc*. 2016;104(3):240-243. doi:10.3163/1536-5050.104.3.014. Erratum in: *J Med Libr Assoc*. 2017;105(1):111.
7. Covidence systematic review software. access date June 27, 2022. Veritas Health Innovation, Melbourne, Australia. www.covidence.org
8. Beniczky S, Ryvlin P. Standards for testing and clinical validation of seizure detection devices. *Epilepsia*. 2018;59(S1):9-13. doi:10.1111/epi.14049
9. Shen YT, Chung PC, Thonnet M, Chauvel P. Seizure detection on prolonged-EEG videos. In: *2008 IEEE International Symposium on Circuits and Systems*; 18-21 May 2008. Seattle, WA: IEEE: 2030-2033. doi:10.1109/ISCAS.2008.4541846
10. Kouamou G, Ferrari G, Lofino F, Raheli R, Pisani F. Extraction of video features for real-time detection of neonatal seizures. In: *2011 IEEE International Symposium on a World of Wireless, Mobile and Multimedia Networks*; 20-24 June 2011. Lucca, Italy: IEEE: 1-6. doi:10.1109/WoWMoM.2011.5986193
11. Lu H, Eng HL, Mandal B, Chan DWS, Ng YL. Markerless video analysis for movement quantification in pediatric epilepsy monitoring. In: *2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society*; 30 August 2011-03 September 2011. Boston, MA: IEEE: 8275-8278. doi:10.1109/IEMBS.2011.6092040
12. Murashov D, Obukhov Y, Kershner I, Sinkin M. Detecting events in video sequence of video-EEG monitoring. *Int Arch Photogramm, Remote Sens Spatial Inf Sci*. 2019;42:155-159. doi:10.5194/isprs-archives-XLII-2-W12-155-2019
13. Ahméd-Aristizabal D, Nguyen K, Denman S, et al. Vision-based mouth motion analysis in epilepsy: a 3D perspective. In: *2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*; 23-27 July 2019. Berlin, Germany: IEEE. 1625-1629. doi:10.1109/EMBC.2019.8857656
14. Cuppens K, Chen CW, Wong KB-Y, et al. Using spatio-temporal interest points (STIP) for myoclonic jerk detection in nocturnal video. In: *2012 Annual International Conference of the IEEE Engineering in Medicine and Biology Society*; 28 August 2012-01 September 2012. San Diego, CA: IEEE: 4454-4457. doi:10.1109/EMBC.2012.6346955
15. Fahmy G, Illiyasu AM, Tawfik A. Automatic seizure detection in face videos based on CWT. In: *2018 IEEE International Symposium on Signal Processing and Information Technology (ISSPIT)*; 06-08 December 2018. Louisville, KY: IEEE: 58-62. doi:10.1109/ISSPIT.2018.8642670
16. Lu H, Pan Y, Mandal B, Eng H-L, Guan C, Chan DWS. Quantifying limb movements in epileptic seizures through color-based video analysis. *IEEE Trans Biomed Eng*. 2013;60(2):461-469. doi:10.1109/TBME.2012.2228649
17. Mandal B, Eng HL, Lu H, Chan DWS, Ng YL. Non-intrusive head movement analysis of videotaped seizures of epileptic

- origin. In: *2012 Annual International Conference of the IEEE Engineering in Medicine and Biology Society*; 28 August 2012-01 September 2012. San Diego, CA: IEEE: 6060-6063. doi:10.1109/EMBC.2012.6347376
18. Murashov DM, Obukhov YV, Kershner IA, Sinkin MV. An algorithm for detecting events in video EEG monitoring data of patients with craniocerebral injuries. *Computer Optics*. 2021; 45(2):301-305. doi:10.18287/2412-6179-CO-798
 19. Ntonfo GMK, Ferrari G, Raheli R, Pisani F. Low-complexity image processing for real-time detection of neonatal clonic seizures. *IEEE Trans Inf Technol Biomed*. 2012;16(3):375-382. doi:10.1109/TITB.2012.2186586
 20. Ntonfo GMK, Lofino F, Ferrari G, Raheli R, Pisani F. Video processing-based detection of neonatal seizures by trajectory features clustering. In: *2012 IEEE International Conference on Communications (ICC)*; 10-15 June 2012. Ottawa, Canada: IEEE: 3456-3460. doi:10.1109/ICC.2012.6364396
 21. Ogura Y, Hayashi H, Nakashima S, et al. A neural network based infant monitoring system to facilitate diagnosis of epileptic seizures. In: *2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*; 5-29 August 2015. Milan, Italy: IEEE: 5614-5617. doi:10.1109/EMBC.2015.7319665
 22. Ojanen P, Knight A, Hakala A, et al. An integrative method to quantitatively detect nocturnal motor seizures. *Epilepsy Res*. 2021;169:106486. doi:10.1016/j.eplepsyres.2020.106486
 23. Pediaditis M, Ciubotaru A-N, Brunschweiler T, et al. Machine learning techniques for personalized detection of epileptic events in clinical video recordings. *AMIA Annu Symp Proc*. 2020;2020:1003-1011.
 24. Pediaditis M, Tsiknakis M, Koumakis L, Karachaliou M, Voutoufiannakis S, Vorgia P. Vision-based absence seizure detection. In: *2012 Annual International Conference of the IEEE Engineering in Medicine and Biology Society*; 28 August 2012-01 September 2012. San Diego, CA: IEEE: 65-68. doi:10.1109/EMBC.2012.6345872
 25. Achilles F, Tombari F, Belagiannis V, Loesch AM, Noachtar S, Navab N. Convolutional neural networks for real-time epileptic seizure detection. *Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization*. 2018;6(3):264-269. doi:10.1080/21681163.2016.1141062
 26. Ahmiedt-Aristizabal D, Denman S, Nguyen K, Sridharan S, Dionisio S, Fookes C. Understanding patients' behavior: vision-based analysis of seizure disorders. *IEEE J Biomed Health Inform*. 2019;23(6):2583-2591. doi:10.1109/JBHI.2019.2895855
 27. Ahmiedt-Aristizabal D, Fookes C, Denman S, et al. A hierarchical multimodal system for motion analysis in patients with epilepsy. *Epilepsy Behav*. 2018;87:46-58. doi:10.1016/j.yebeh.2018.07.028
 28. Cattani L, Alinovi D, Ferrari G, et al. Monitoring infants by automatic video processing: a unified approach to motion analysis. *Comput Biol Med*. 2017;80:158-165. doi:10.1016/j.compbiomed.2016.11.010
 29. Cattani L, Kouamou Ntonfo GM, Lofino F, Ferrari G, Raheli R, Pisani F. "Maximum-likelihood detection of neonatal clonic seizures by video image processing," In: *2014 8th International Symposium on Medical Information and Communication Technology (ISMICT)*; 02-04 April 2014. Firenze, Italy: IEEE: 1-5. doi:10.1109/ISMICT.2014.6825219
 30. Geertsema EE, Thijs RD, Gutter T, et al. Automated video-based detection of nocturnal convulsive seizures in a residential care setting. *Epilepsia*. 2018;59(Suppl 1):53-60. doi:10.1111/epi.14050
 31. Kalitzin S, Petkov G, Velis D, Vledder B, Lopes Da Silva F. Automatic segmentation of episodes containing epileptic clonic seizures in video sequences. *IEEE Transactions on Biomedical Engineering*. 2012;59(12):3379-3385. doi:10.1109/TBME.2012.2215609
 32. Kalitzin SN, Bauer PR, Lamberts RJ, Velis DN, Thijs RD, Da Silva FHL. Automated video detection of epileptic convulsion slowing as a precursor for post-ictal generalized EEG suppression. In: Kyriacou E, Christofides S, Pattichis CS, eds. *XIV Mediterranean Conference on Medical and Biological Engineering and Computing*. Springer International Publishing; 2016:57:100-104. doi:10.1007/978-3-319-32703-7_21
 33. Karayiannis NB, Tao G, Frost JD, Wise MS, Hrachovy RA, Mizrahi EM. Automated detection of videotaped neonatal seizures based on motion segmentation methods. *Clin Neurophysiol*. 2006;117(7):1585-1594. doi:10.1016/j.clinph.2005.12.030
 34. Karayiannis NB, Xiong Y, Frost JD, Wise MS, Hrachovy RA, Mizrahi EM. Automated detection of videotaped neonatal seizures based on motion tracking methods. *J Clin Neurophysiol*. 2006;23(6):521-531. doi:10.1097/00004691-200612000-00004
 35. Van Westrhenen A, Petkov G, Kalitzin SN, Lazerion RHC, Thijs RD. Automated video-based detection of nocturnal motor seizures in children. *Epilepsia*. 2020;61(S1):S36-S40. doi:10.1111/epi.16504
 36. Xiong YH, Karayiannis NB. Automated recognition of videotaped neonatal seizures using robust motion tracking methods that adjust to illumination and contrast changes. In: *3rd IEEE International Symposium on Biomedical Imaging: Macro to Nano*, 2006; 06-09 April 2006. Arlington, VA: IEEE: 1300-1303. doi:10.1109/ISBI.2006.1625164
 37. Yang Y, Sarkis RA, Atrache RE, Loddenkemper T, Meisel C. Video-based detection of generalized tonic-clonic seizures using deep learning. *IEEE J Biomed Health Inform*. 2021;25(8):2997-3008. doi:10.1109/JBHI.2021.3049649
 38. Peltola J, Basnyat P, Armand Larsen S, et al. Semiautomated classification of nocturnal seizures using video recordings. *Epilepsia*. 2022;64(Suppl 4):S65-S71. doi:10.1111/epi.17207
 39. Pisani F, Spagnoli C, Pavlidis E, et al. Real-time automated detection of clonic seizures in newborns. *Clin Neurophysiol*. 2014; 125(8):1533-1540. doi:10.1016/j.clinph.2013.12.119
 40. Basnyat P, Mäkinen J, Saarinen JT, Peltola J. Clinical utility of a video/audio-based epilepsy monitoring system Nelli. *Epilepsy Behav*. 2022;133:108804. doi:10.1016/j.yebeh.2022.108804
 41. Ahmiedt-Aristizabal D, Fookes C, Nguyen K, Denman S, Sridharan S, Dionisio S. Deep facial analysis: a new phase I epilepsy evaluation using computer vision. *Epilepsy Behav*. 2018;82:17-24. doi:10.1016/j.yebeh.2018.02.010
 42. Fang Z, Leung H, Choy CS. Spatial temporal GRU convnets for vision-based real time epileptic seizure detection. In: *2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018)*; 04-07 April 2018. Washington, DC: IEEE: 1026-1029. doi:10.1109/ISBI.2018.8363746